

CS339: Abstractions and Paradigms for Programming

Environment Model

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Problems brought by assignments

```
(define count 1)
(define (f v)
  (set! count (+ 1 count))
  (+ v count))
```

```
> (f 2)
> 4
> (f 2)
> 5
```

- Procedures are no longer mathematical functions!

```
(define (make-simplified-withdraw balance)
  (lambda (amount)
    (set! balance (- balance amount))
    balance))
```

```
((make-simplified-withdraw 25) 20)
==> ((lambda (amount) (set! balance (- 25 amount) 25) 20)
==> ((set! balance (- 25 20)) 25)
==> Set balance to 5 and return 25
```

- Substitution model of procedure application no longer works!
 - Reason: Variables are no longer simply names for expressions!



Substitution model revisited

- **Evaluating a combination:**

- Evaluate the subexpressions of the combination.
- Apply the value of the operator subexpression to the values of the operand subexpressions.

- **Applying a procedure:**

- To apply a compound procedure to arguments, evaluate the body of the procedure with each formal parameter replaced by the corresponding actual argument.



Bindings and scope revisited

This lambda
binds 'x'

This lambda
binds 'y'

```
(lambda (x) (lambda (y) (* x y)))
```

Scope of y

Scope of x

```
(lambda (x) (lambda (y) (* x (+ z y))))
```

x and y are bound;
z is free

The only way to create a binding is via a lambda!



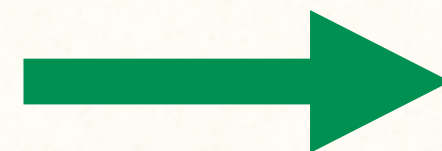
Lambda — The Ultimate Binder

```
(let ((var1 expr1)
      (var2 expr2))
  <body>)
```



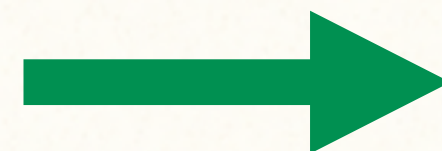
```
((lambda (var1 var2)
  <body>)
  expr1 expr2)
```

```
(define (foo x)
  <body>)
```



```
(define foo
  (lambda (x)
    <body>))
```

```
(define (foo x)
  (define (var expr)
    <body>)...)
```



```
(define (foo x)
  (let ((var expr))
    <body>)...)
```

The Environment Model

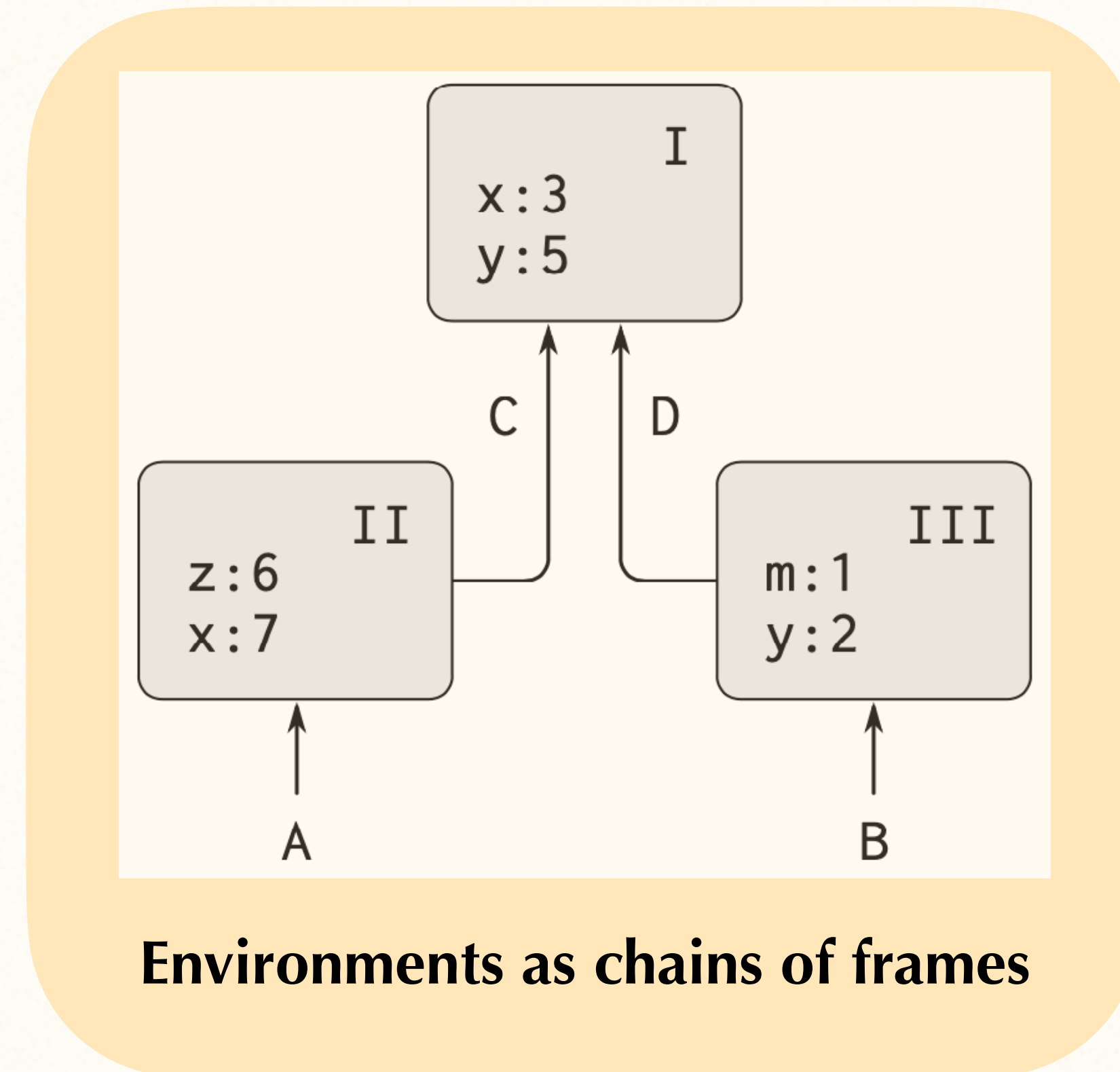


Frames and environments

I, II, III: Frames

Environment 'A':

- x: 7, y: 5, z: 6
- m: unbound



Environments as chains of frames

Environments 'C', 'D':

- x: 3, y: 5
- z, m: unbound

A, B, C, D: Environments

Environment 'B':

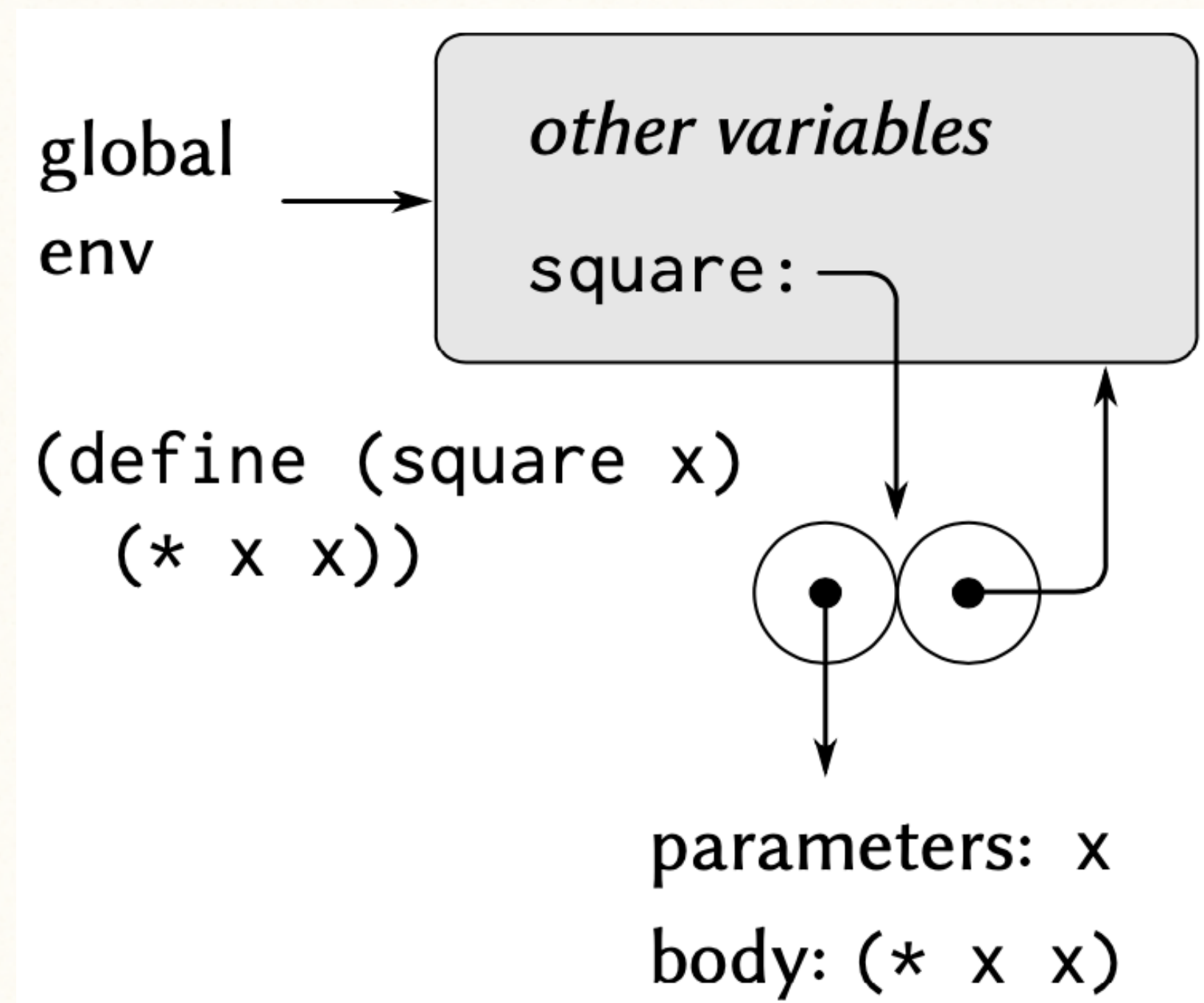
- x: 3, y: 2, m: 1
- z: unbound

Evaluating procedure definitions

```
(define (square x)  
  (* x x))
```



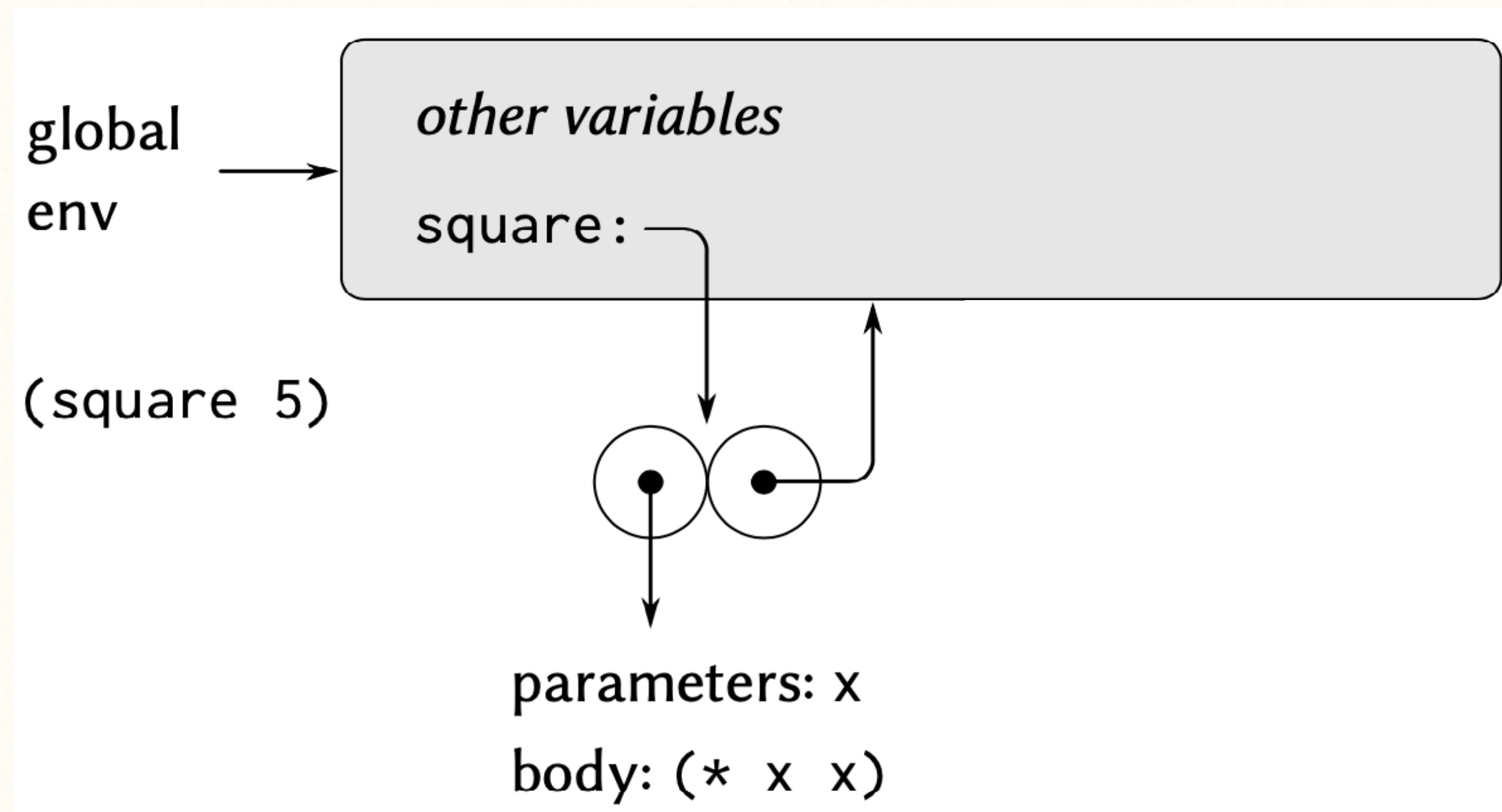
```
(define square  
  (lambda (x) (* x x)))
```



Evaluating procedure applications

```
(define (square x)  
  (* x x))
```

```
> (square 5)
```

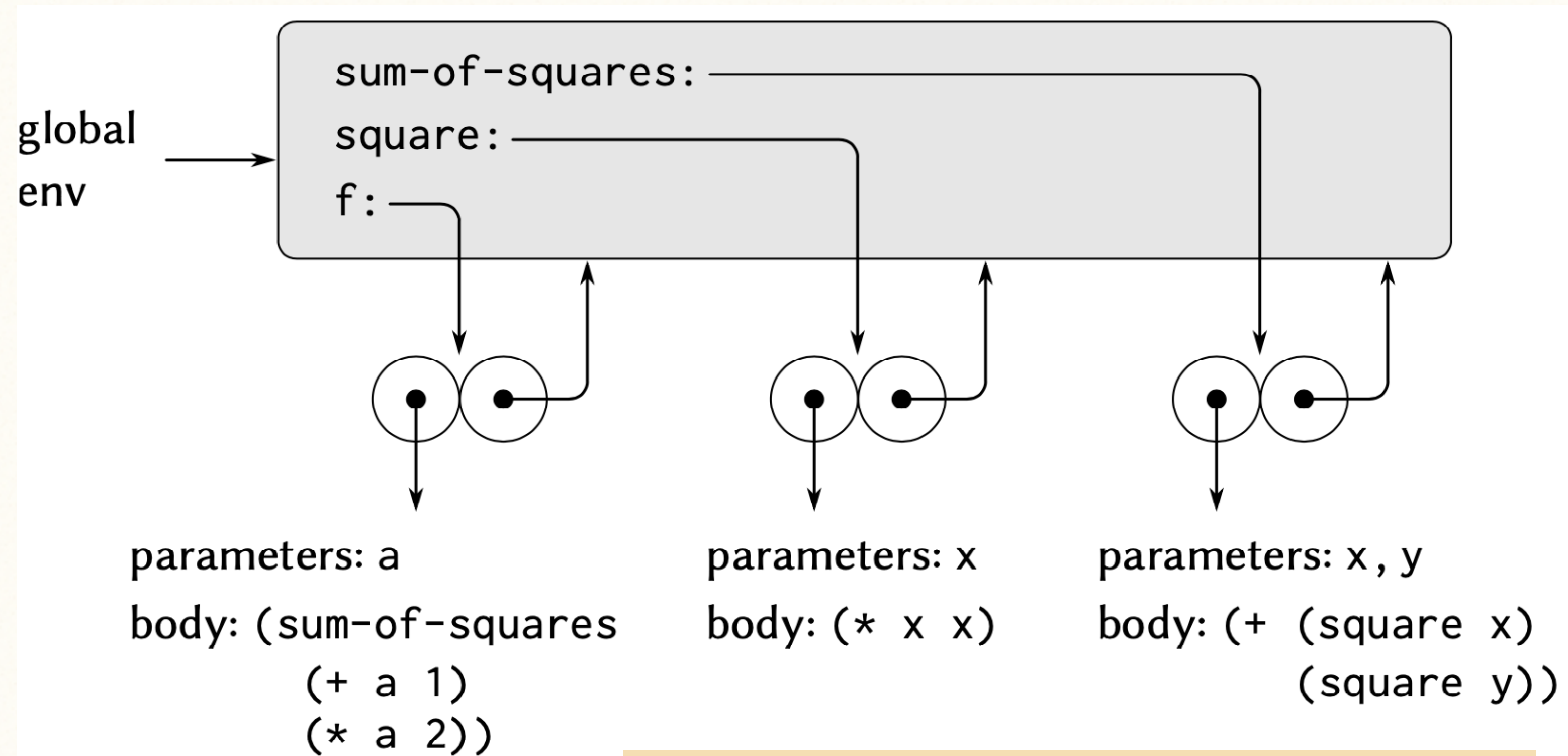


The Environment Model: Summary

- **Evaluating a combination:** Same as the substitution model
 - Evaluate the subexpressions of the combination.
 - Apply the value of the operator subexpression to the values of the operand subexpressions.
- **Evaluating a procedure definition:**
 - Create a procedure object by evaluating a lambda-expression relative to a given environment.
- **Evaluating a procedure application:**
 - Create a new environment containing a frame that binds the parameters to the arguments, and then evaluate the body of the procedure **in the** new environment.

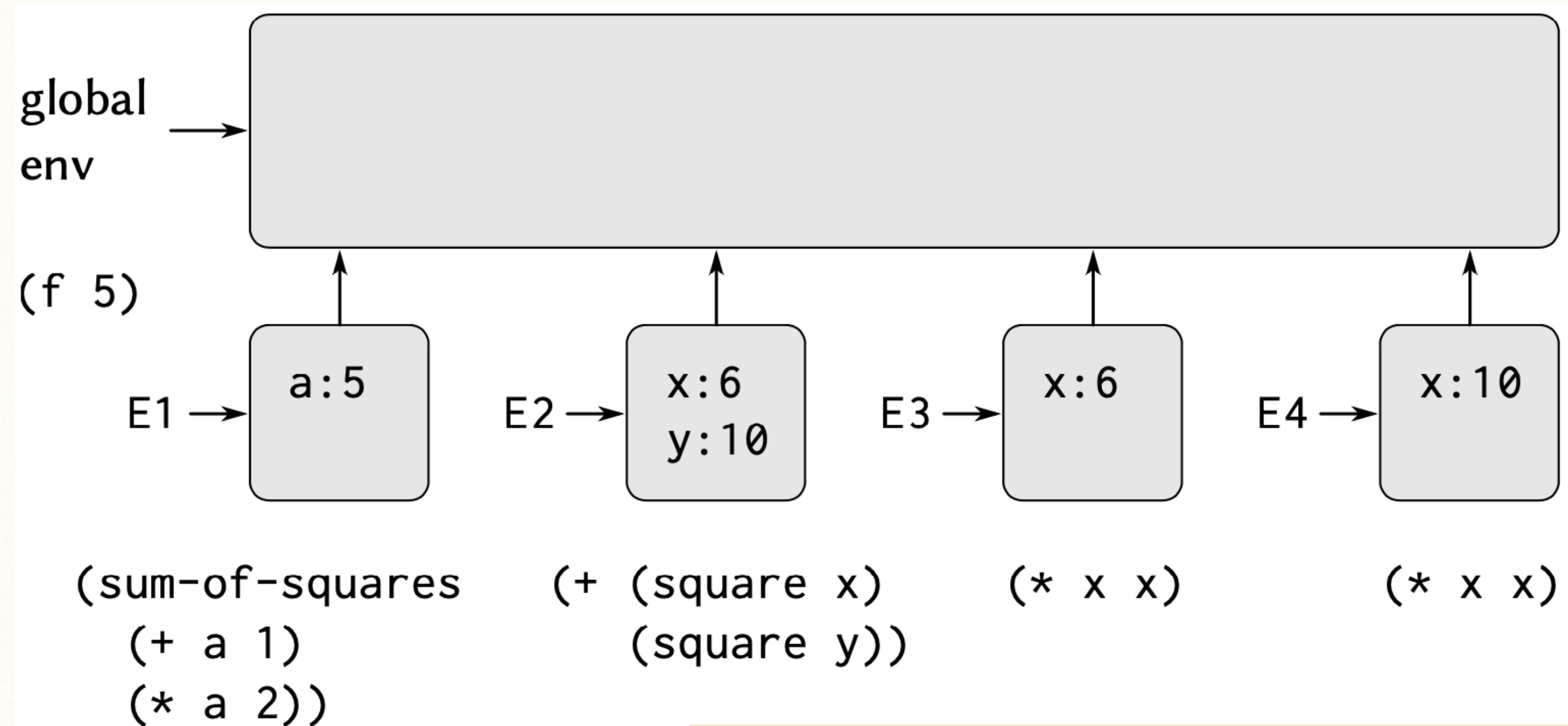


Extended example 1: Definition



```
(define (square x)
  (* x x))
(define (sum-of-squares x y)
  (+ (square x) (square y)))
(define (f a)
  (sum-of-squares (+ a 1) (* a 2)))
```


Extended example 1: Application

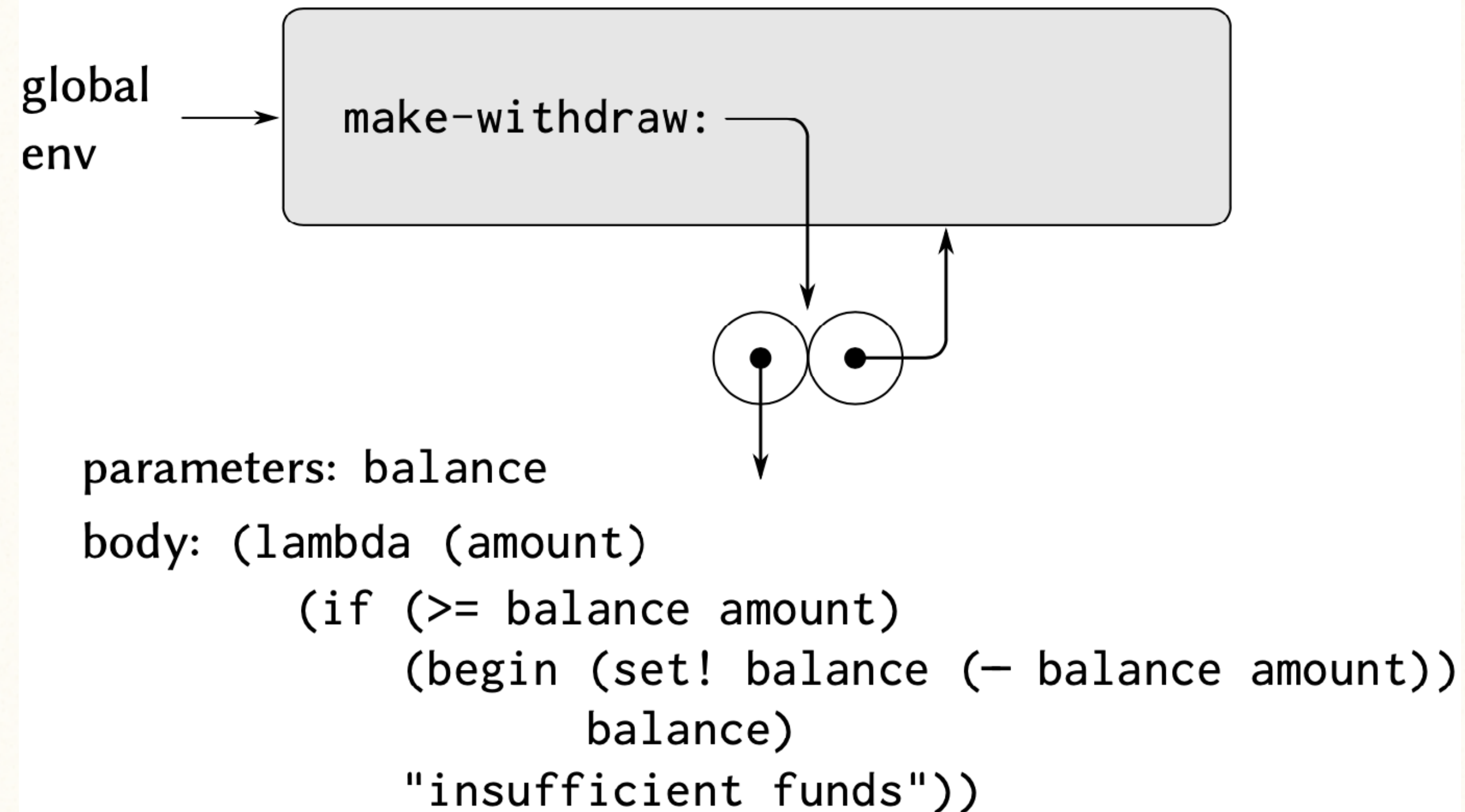


```
(define (square x)
  (* x x))
(define (sum-of-squares x y)
  (+ (square x) (square y)))
(define (f a)
  (sum-of-squares (+ a 1) (* a 2)))

> (f 5)
```



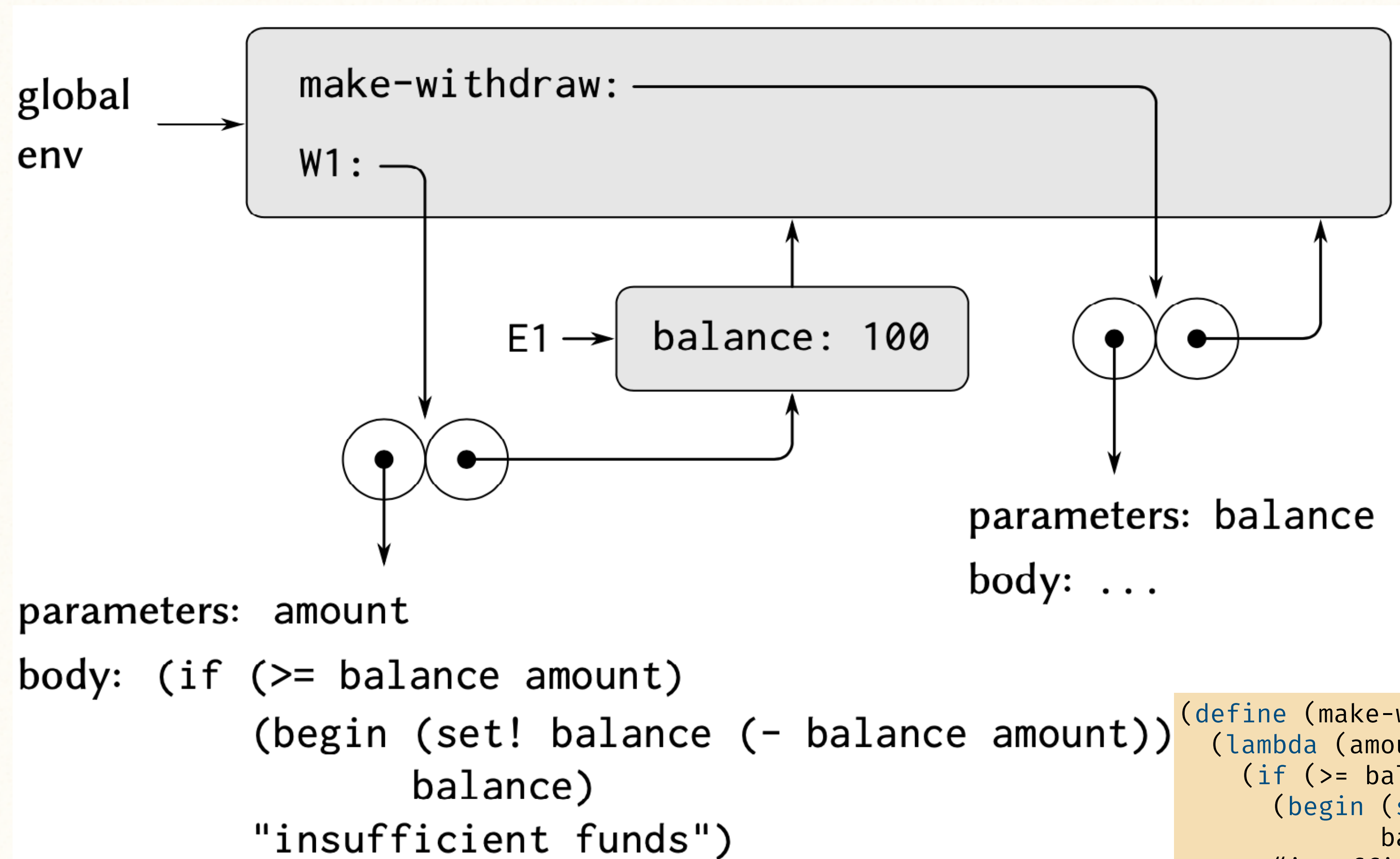
EE2 Def: Local states in frames



```
(define (make-withdraw balance)  
  (lambda (amount)  
    (if (>= balance amount)  
        (begin (set! balance (- balance amount))  
                 balance)  
        "insufficient funds")))
```



EE2 App1

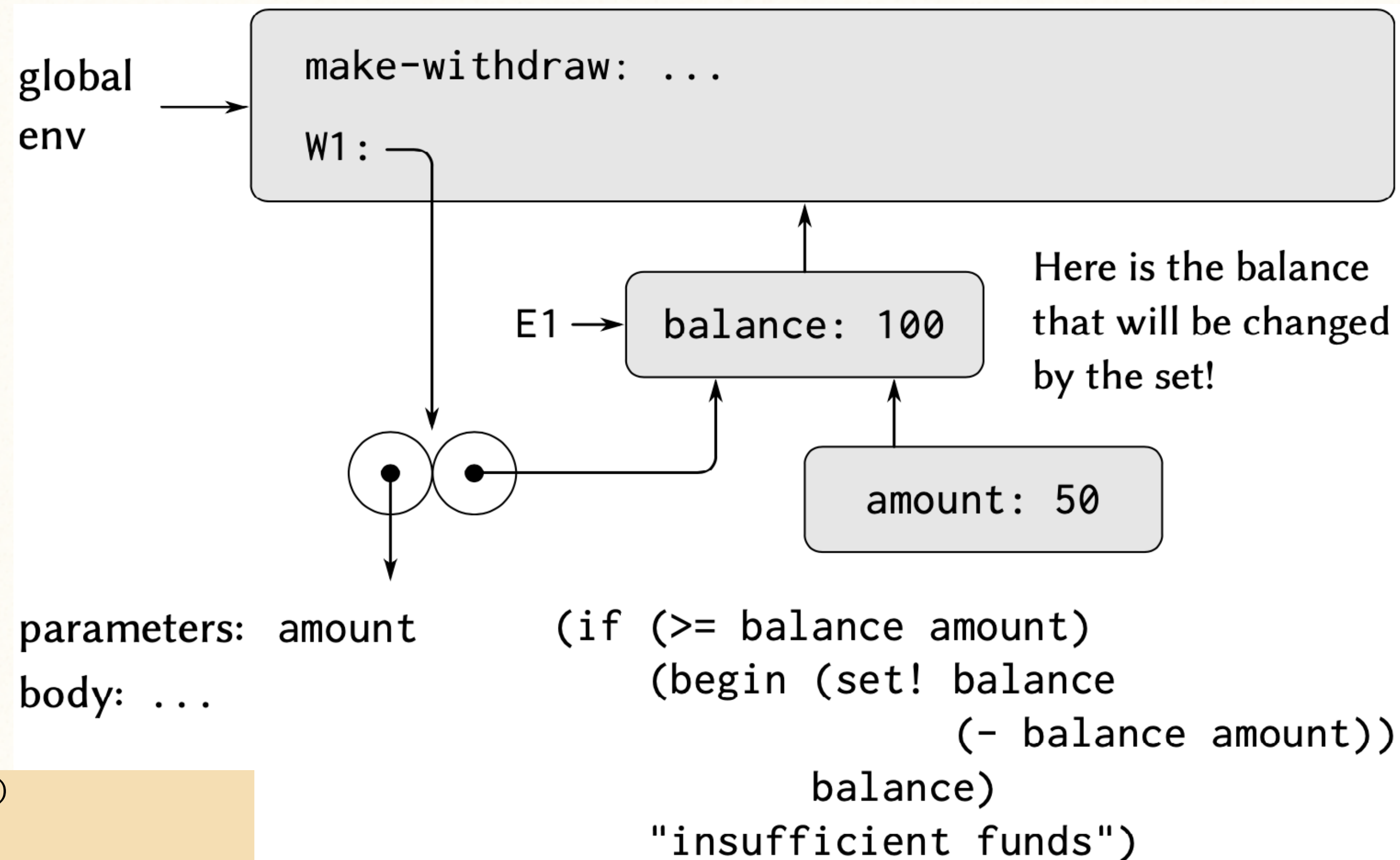


```
(define (make-withdraw balance)
  (lambda (amount)
    (if (>= balance amount)
        (begin (set! balance (- balance amount))
                balance)
        "insufficient funds"))))
```

```
(define W1 (make-withdraw 100))
```



EE2 App1 (Cont.)



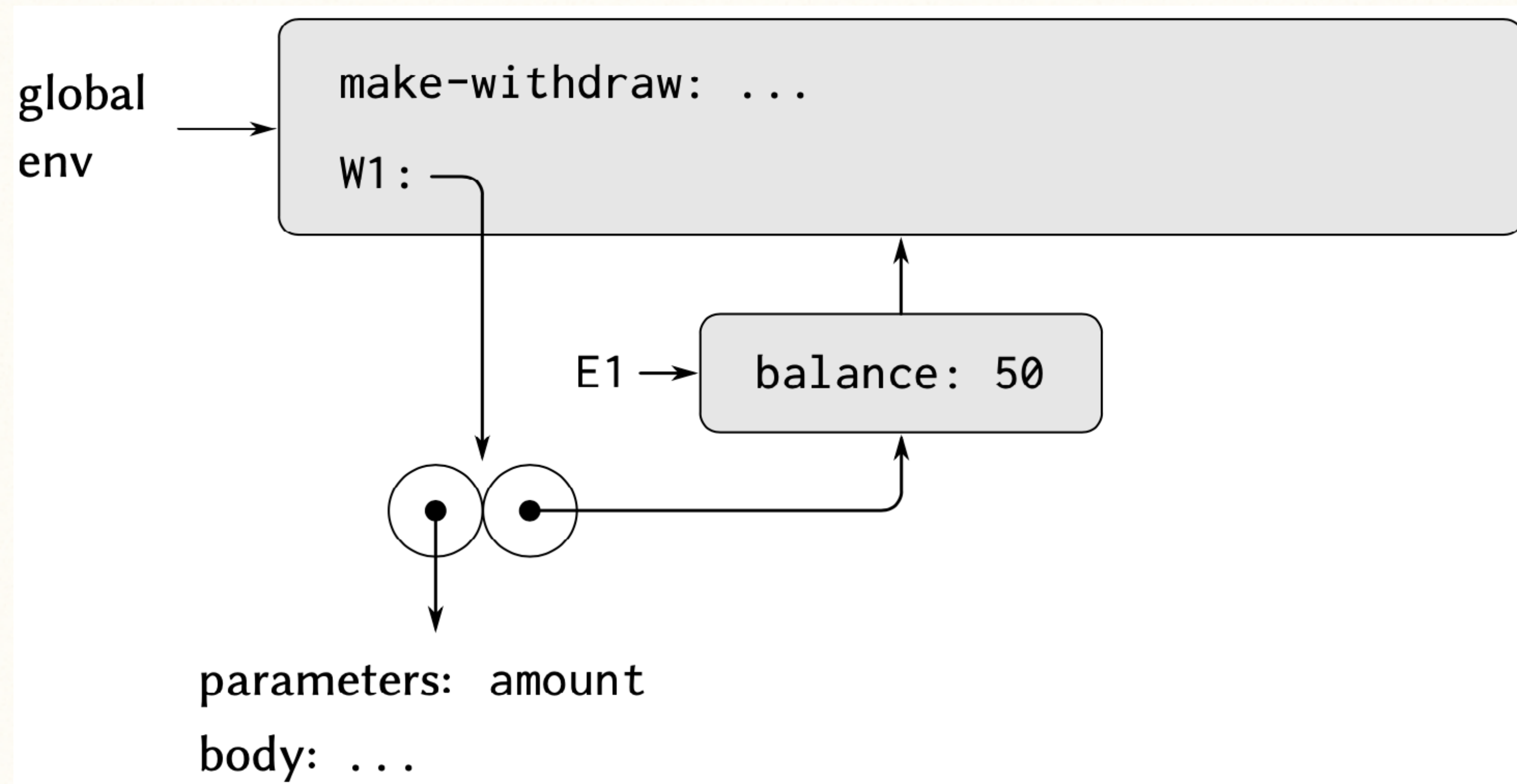
```
(define (make-withdraw balance)
  (lambda (amount)
    (if (>= balance amount)
        (begin (set! balance (- balance amount))
                balance)
        "insufficient funds")))
```

```
(define W1 (make-withdraw 100))
```

```
> (W1 50)
```

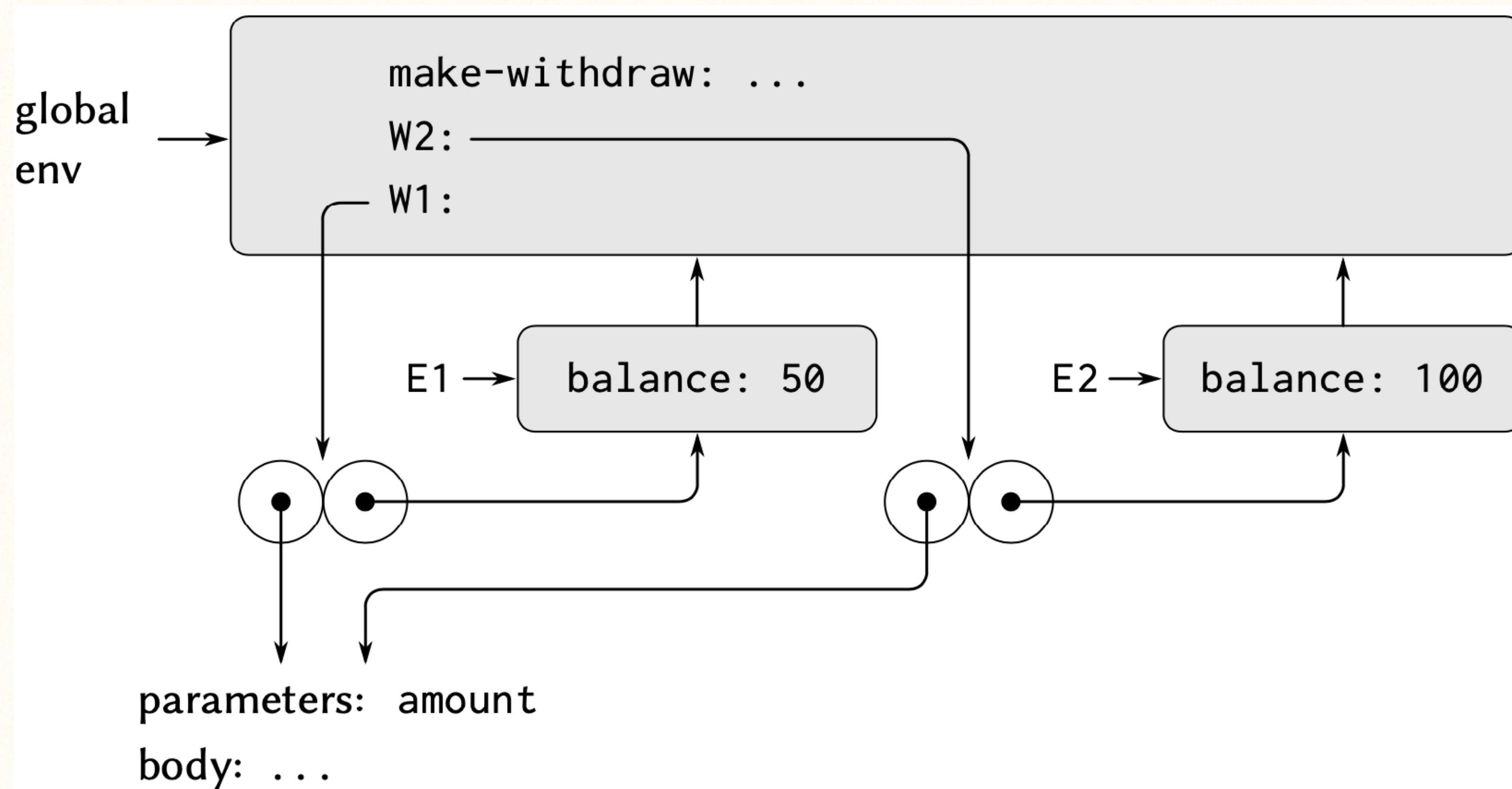


EE2 App1 (Cont.)



After (W1 50)

EE2 App2

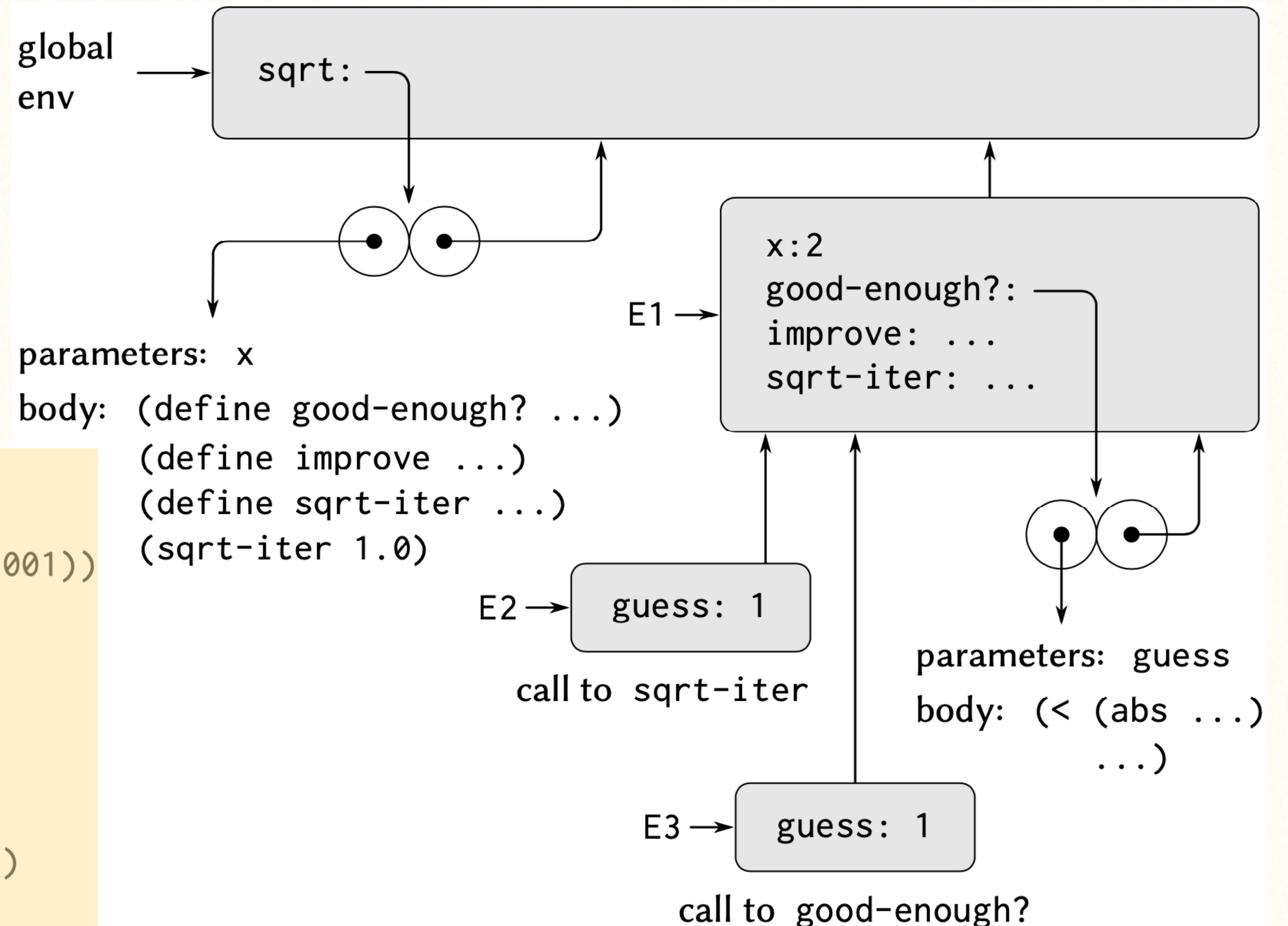


```
(define W2 (make-withdraw 100))
```


Internal Definitions

> (sqrt 2)

```
(define (sqrt x)
  (define (good-enough? guess)
    (< (abs (- (square guess) x)) 0.001))
  (define (improve guess)
    (average guess (/ x guess)))
  (define (sqrt-iter guess)
    (if (good-enough? guess)
        guess
        (sqrt-iter (improve guess))))
  (sqrt-iter 1.0))
```



Unevaluated...



New syntax, same ghost
— the environment model
didn't disappear. It just put on a
suit and called itself a class.